The Cascade Curvature Operator under Refinement: Schur–Complement Identity and Energy Intertwining

A predecessor to the cascade-closure mechanism paper

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Abstract

We isolate a clean abstract model — a scalar vertex-sector graph Laplacian under pure edge-midpoint subdivision with degree-2 midpoints — and prove a Schur-complement halving identity in this model that gives an abstract-model curvature-level analogue of the source's coboundary factor-$\frac{1}{2}$ relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ from [3]. With the scaled vertex Laplacian $\widetilde{A}_n := 2^n A_n^{\text{vertex}}$, the Schur complement under the (boundary, interior) partition equals the previous-level operator exactly: $\text{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$. The identity is verified analytically and numerically on two concrete instances of the abstract model (single edge with one-midpoint subdivision; triangle with edge-midpoint subdivision) and yields the cascade-mechanism residual-monotonicity hypothesis $\widetilde{R}_n(p^0\psi) \le \widetilde{R}_{n+1}(\psi)$ for every $\psi \in X^0_{n+1}$, with equality iff ψ is the harmonic extension of $p^0\psi$. Strict operator-level refinement-compatibility $p^0\widetilde{A}_{n+1} = \widetilde{A}_n p^0$ remains false; the defect operator vanishes on harmonic extensions and is generically non-zero on the level-n-supported sector $\{\psi : \psi_I = 0\}$. We *do not* extend the result to the full Schläfli-refinement P3/P4 tower of [4, 3] (which adds face-centroids, midpoint-centroid edges, and F-valued chain spaces with antisymmetric oriented-edge convention); the gap between this paper's abstract scalar pure-midpoint model and that fuller setting is named explicitly in §1.1 as the remaining upstream task. The result discharges cascade-mechanism's obligation (O3) for the abstract model and discharges a boundary-energy / Schur substitute for (O2) in the abstract model ((O2) as stated, at the operator/flow level, remains false). The result recasts the cascade-mechanism worked-invocation status table accordingly.

Scope

- This is a narrow technical paper on an *abstract scalar model* of refinement (vertex-sector graph Laplacian, pure edge-midpoint subdivision with degree-2 midpoints), not the full Schläfli-refinement P3/P4 tower of [4, 3] (see §1.1). It contains no new mechanism, no programme framing, no consciousness or cosmology claim. It is the source-side companion to [2], applied to the abstract model.
- Within the abstract scalar model: (O0) closed; (O3) closed via the Schur halving identity (Theorem 2.1, Theorem 3.1); (O2) *as stated* (operator / flow level) remains false (Proposition 4.1); the boundary-energy / Schur substitute for (O2) is discharged; (O1) recorded but not nontrivially discharged (Proposition 5.1).
- Lifting these results to the full P3/P4 tower (centroids, midpoint-centroid extra edges, F-valued X_n^0 and antisymmetric X_n^1 chain spaces) is named as the remaining upstream task and is not attempted here.

1 Introduction and source-state recap

The cascade-closure mechanism paper [2] defines cascade closure as a rung-indexed closure-residual / projection-stack event, proves a conditional residual-zero / fixed-point propagation proposition, and supplies a consumer API listing four obligations that successor papers must discharge to invoke cascade closure as a load-bearing premise:

(O0) Standing analytic / admissibility hypotheses.

(O1) Top-rung fixed point and convergence.

(O2) Flow intertwining $\pi^{\sharp}_{k+1,k} \circ \Psi^t_{k+1} = \Psi^t_k \circ \pi^{\sharp}_{k+1,k}$.

(O3) Residual monotonicity $R_k(\pi^{\sharp}_{k+1,k}\psi) \leq R_{k+1}(\psi)$.

The original source-state analysis recorded in [2] §3.5 (the open-state pre-incorporation paragraph, before the abstract-model update of the present paper was folded back) established the per-obligation status in the imported P3/P4 finite-level tower of [4, 3]: (O0) is satisfied at the finite-dimensional level; (O1)–(O3) are open in the source state, with (O2) closed only for the source's mass-only operator $L_n = -\gamma C_n$ at $\alpha = \beta = 0$, not for the curvature/residual operator A_n that the cascade-mechanism flow naturally uses. [2] §3.5 names the minimal upstream upgrade required and observes that the role of [2] is to make the obligations visible as the programme's actual dependency gate, not to discharge them. ([2] §3.5 already incorporates this paper's abstract-model update into the worked-invocation status in its later paragraphs; the present paper is the source-side companion that supplies that update.)

The present paper is a partial source-side discharge, carried out in a deliberately scaled-down setting: an *abstract scalar pure-midpoint refinement model* (Definition 1.1 below) that retains the load-bearing coboundary refinement relation of the imported tower while dropping the centroids, midpoint-centroid edges, and F-valued / antisymmetric oriented-edge structure of the full Schläfli refinement. In this abstract model, with the curvature operator A_n rather than the source's mass-only L_n, we supply a single source-side identity and exhibit its consequences for (O2) and (O3). The lift to the full P3/P4 tower of [4, 3] is named explicitly in §1.1 as hypotheses (L1)–(L3) and is not attempted here. We do not redefine the cascade-closure mechanism, do not modify [2]'s consumer API, and do not introduce any programme-level claim.

1.1 Scope: abstract scalar pure-midpoint model vs. full P3/P4 tower

The actual source refinement of [4] (Definition `def:refinement`) is the *Schläfli refinement*: at each level n it adds, alongside the new edge-midpoints $M_n^\bullet$, a separate set of face-centroids $C_n^\bullet$ (one per 2-face of the underlying 4-polytope), together with extra level-$(n+1)$ edges connecting midpoints to centroids and centroids to other midpoints in a $W^\bullet$-equivariant pattern. The chain spaces in the source are also F-valued (Definitions `def:Xn-0` and `def:Xn-1`): $X_n^{0,\bullet}$ is F^0-valued on $V(G_n^\bullet)$ and $X_n^{1,\bullet}$ is F^1-valued and antisymmetric on *oriented* edges, with edge-parent map $\pi^E_{n+1,n}$ taking the value $\perp$ on the new midpoint-centroid, centroid-midpoint, and (where present) midpoint-midpoint edges of [4].

The present paper does *not* work in this fuller setting. We work in a deliberate *abstract scalar pure-midpoint model* defined as follows:

Definition 1.1 (Abstract scalar pure-midpoint refinement model)**.** A pair of finite, simple, undirected graphs $G_n = (V_n, E_n)$ and $G_{n+1} = (V_{n+1}, E_{n+1})$ (no loops, no parallel edges) together with bonding maps is in the abstract scalar pure-midpoint model when:

(M1) $V_{n+1} = V_n \sqcup M_n$, where M_n contains exactly one new *midpoint* vertex m_e per edge $e \in E_n$, and no other new vertices (no centroids);

(M2) each level-n edge $e \in E_n$ is given a fixed orientation $e = (u, v)$, and the level-$(n+1)$ child edges inherit it coherently: $E_{n+1} = \{(u, m_e), (m_e, v) : e = (u, v) \in E_n\}$, so that traversing both children in order recovers the parent orientation, and there are no extra midpoint–midpoint edges;

(M3) the chain spaces are scalar: $X_n^0 = \mathbb{R}^{V_n}$, $X_n^1 = \mathbb{R}^{E_n}$, with standard ℓ^2 inner products. Edge chains are oriented per (M2): X_n^1 uses the fixed orientation of E_n;

(M4) bonding maps are vertex-restriction $p^0_{n+1,n} : X^0_{n+1} \to X^0_n$ along the inclusion $V_n \subset V_{n+1}$ and parent-fibre average $p^1_{n+1,n} : X^1_{n+1} \to X^1_n$ by factor $\frac{1}{2}$ on each of the two coherently oriented children of every parent edge.

The vertex-sector curvature is the standard graph Laplacian $A_n^{\text{vertex}} := d_n^* d_n$ on X_n^0, with $d_n : X_n^0 \to X_n^1$ the oriented difference under (M2), $(d_n f)(u, v) = f(v) - f(u)$.

In this abstract model, the coboundary refinement relation

$$p^1_{n+1,n} d_{n+1} = \tfrac{1}{2}\, d_n\, p^0_{n+1,n} \tag{1}$$

holds verbatim, by direct computation: for any $f \in X^0_{n+1}$, $(p^1 d_{n+1} f)(e) = \frac{1}{2}[(d_{n+1}f)(u, m_e) + (d_{n+1}f)(m_e, v)] = \frac{1}{2}[(f(m_e) - f(u)) + (f(v) - f(m_e))] = \frac{1}{2}(f(v) - f(u)) = \frac{1}{2}(d_n p^0 f)(e)$. We use (1) as the load-bearing input below. The relation also holds in the source's full Schläfli setting ([3] Proposition `prop:coboundary-refinement`), but the present paper only uses the abstract-model version, where it is elementary.

The gap to the full P3/P4 tower. The theorems, propositions, and corollary below all hold only *within Definition 1.1's abstract model.* They do *not* immediately transfer to the full Schläfli setting of [4]. The sufficient lift hypotheses we isolate are the following two open conditions, plus one already-proved input:

(L1) (open) the level-$(n + 1)$ vertex Laplacian, restricted to boundary vertices $V_n \subset V(G^\bullet_{n+1})$, agrees with the Laplacian of the abstract model up to a defect that is supported only on midpoints (no extra contribution from centroid neighbours or centroid-midpoint edges);

(L2) (already-proved input) the parent-fibre averaging map $p^1_{n+1,n}$ on F^1-valued, antisymmetric oriented-edge chains satisfies the calibration identity $p^1(p^1)^* = \frac{1}{2}\, I$ in the F^1 inner product, with the same factor of $\frac{1}{2}$ as in the scalar abstract model. This is [4] Proposition `prop:adjoints`, parts (3)–(4); we retain the label here for downstream reference;

(L3) (open) the harmonic-extension operator H_{n+1} in the F^0-valued setting saturates the energy lower bound coming from (L1) and (L2).

Each of (L1) and (L3) is concrete and isolatable: (L1) is a graph-theoretic incidence statement about the $W^\bullet$-equivariant midpoint and centroid placement; (L3) is the analogue of Step 3 of the proof of Theorem 2.1 in the F-valued setting. (L2) is recorded for completeness; it does not need re-proof. Lifting the Schur-halving identity to the full P3/P4 tower under these conditions is named here as the remaining upstream task and is not attempted in this paper.

1.2 Source objects (recap)

Within the abstract model of Definition 1.1, we use scalar finite-dimensional chain spaces. The vertex-block curvature operator is

$$A_n^{\text{vertex}} := d_n^* d_n \in \mathbb{R}^{V_n \times V_n}, \tag{2}$$

the standard graph Laplacian on V_n.

The bonding maps are the pair $p_{n+1,n} = (p^0_{n+1,n}, p^1_{n+1,n})$, where $p^0_{n+1,n} : X^0_{n+1} \to X^0_n$ is vertex-restriction along the canonical inclusion $V_n \subset V_{n+1}$ and $p^1_{n+1,n} : X^1_{n+1} \to X^1_n$ is the parent-fibre average by factor $\frac{1}{2}$ on each of the two children of every parent edge.

The coboundary refinement relation (1) of §1.1 is the load-bearing input below. (The analogous identity is also a theorem in the source's full Schläfli setting, [3] Proposition `prop:coboundary-refinement`.)

1.3 What this paper proves

The paper's central object is the scaled vertex curvature operator:

Definition 1.2 (Scaled curvature operator)**.** $\widetilde{A}_n := 2^n A_n^{\text{vertex}}$, defined on X^0_n. The corresponding scaled residual is $\widetilde{R}_n(\phi) := \frac{1}{2}\langle \phi, \widetilde{A}_n \phi\rangle = 2^{n-1}\langle \phi, A_n^{\text{vertex}}\phi\rangle$.

For nontrivial refinements (those with at least one parent edge, so $A_n^{\text{vertex}} \neq 0$), the scaling factor 2^n is the unique exponential factor (up to an overall normalisation c, giving $c \cdot 2^n$) for which Theorem 2.1 below produces the Schur-halving identity $\mathrm{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$. (The coboundary relation (1) alone does not fix this scaling; it is forced only after the Schur computation: writing $\widetilde{A}_n = c_n A_n^{\text{vertex}}$ with $c_n > 0$, the identity $\mathrm{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$ together with $\mathrm{Schur}(A_{n+1}^{\text{vertex}}) = \frac{1}{2}A_n^{\text{vertex}}$ forces $c_{n+1}/2 = c_n$, hence $c_n = c_0 \cdot 2^n$.) We fix $c_0 = 1$, equivalently $\widetilde{A}_0 = A_0^{\text{vertex}}$. The main results are an exact Schur-complement identity for $\widetilde{A}_n$ under boundary–interior decomposition (Theorem 2.1), a clean energy-intertwining inequality $\widetilde{R}_n(p^0\psi) \leq \widetilde{R}_{n+1}(\psi)$ for every ψ, with equality *iff* ψ is a harmonic extension, that discharges cascade-mechanism's (O3) (Theorem 3.1), and an explicit defect identity showing that strict operator-level refinement-compatibility for $\widetilde{A}_n$ remains false (Proposition 4.1).

2 Schur-complement identity

Throughout this section and the rest of the paper, we write $G_n^\bullet$ for the abstract-model graph G_n of Definition 1.1 when emphasising the visual continuity with the source-level notation of [4, 3]; the bullet decoration carries no extra structure beyond the abstract-model G_n and is purely a notational convenience.

Fix a level n and a finite refinement $V(G_n^\bullet) \subset V(G_{n+1}^\bullet)$ such that $|V(G_{n+1}^\bullet)| - |V(G_n^\bullet)|$ new "midpoint" vertices are added (each created as the midpoint of one existing edge of $G_n^\bullet$ that was subdivided to produce $G_{n+1}^\bullet$). Partition the vertex set $V(G_{n+1}^\bullet) = B \sqcup I$ into

- $B := V(G_n^\bullet)$, the *boundary* vertices (those retained from level n);
- $I := V(G_{n+1}^\bullet) \setminus V(G_n^\bullet)$, the *interior* or *new* vertices.

Block-decompose A_{n+1}^{vertex} in this (B, I) ordering:

$$A_{n+1}^{\text{vertex}} = \begin{pmatrix} A_{BB} & A_{BI} \\ A_{IB} & A_{II} \end{pmatrix}, \tag{3}$$

where $A_{BB} \in \mathbb{R}^{B\times B}$, $A_{BI} = A_{IB}^T \in \mathbb{R}^{B\times I}$, $A_{II} \in \mathbb{R}^{I\times I}$.

By construction A_n^{vertex} is the graph Laplacian on $V(G_n^\bullet)$. The harmonic extension $H_{n+1} : X_n^0 \to X_{n+1}^0$ is the unique linear map sending $f \in X_n^0$ to the level-$(n+1)$ state with prescribed boundary values f and interior values $f_I = -A_{II}^{-1} A_{IB} f$, equivalently the unique minimiser of the level-$(n+1)$ energy $\frac{1}{2}\langle \cdot\,, A_{n+1}^{\text{vertex}} \cdot\rangle$ subject to the boundary constraint $p_{n+1,n}^0 \cdot = f$. In the abstract model of Definition 1.1, midpoints have level-$(n+1)$ degree exactly 2, so $A_{II} = 2I$ on the midpoint block, hence trivially invertible.

The Schur complement of A_{n+1}^{vertex} with respect to the partition (B, I) is

$$\text{Schur}(A_{n+1}^{\text{vertex}}) := A_{BB} - A_{BI} A_{II}^{-1} A_{IB} \in \mathbb{R}^{B\times B}. \tag{4}$$

Theorem 2.1 (Schur–complement halving in the abstract model)**.** *In the abstract scalar pure-midpoint refinement model of Definition 1.1, with vertex-sector curvature $A_n^{\text{vertex}} = d_n^* d_n$, the Schur complement of A_{n+1}^{vertex} under the (boundary, interior) partition (3) is exactly half the level-n Laplacian:*

$$\text{Schur}(A_{n+1}^{\text{vertex}}) = \tfrac{1}{2}\, A_n^{\text{vertex}}. \tag{5}$$

Equivalently, with $\widetilde{A}_n := 2^n A_n^{\text{vertex}}$:

$$\text{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n. \tag{6}$$

Proof. We proceed in three steps: identify the Schur complement with a harmonic energy, decompose that energy via the source's coboundary refinement relation, and verify directly that the harmonic extension saturates the resulting lower bound.

Step 1. For any positive-semi-definite A_{n+1}^{vertex} with invertible interior block A_{II}, the Schur complement and the harmonic-extension energy agree: $\langle f, \text{Schur}(A_{n+1}^{\text{vertex}}) f\rangle = \min_{\psi : p^0\psi = f} \langle \psi, A_{n+1}^{\text{vertex}} \psi\rangle = \langle H_{n+1} f, A_{n+1}^{\text{vertex}} H_{n+1} f\rangle$ for every $f \in X_n^0$. This is standard: for $\psi = (f, f_I)$ in block form, $\langle \psi, A_{n+1}^{\text{vertex}}\psi\rangle = \langle f, A_{BB} f\rangle + 2\langle f_I, A_{IB} f\rangle + \langle f_I, A_{II} f_I\rangle$ is minimised in f_I at $f_I^\star = -A_{II}^{-1} A_{IB} f$ (since A_{II} is positive definite), with minimum value $\langle f, (A_{BB} - A_{BI} A_{II}^{-1} A_{IB}) f\rangle$, and $H_{n+1} f := (f, f_I^\star)$ by definition of harmonic extension.

Step 2. Let $f \in X_n^0$ and let $\psi \in X_{n+1}^0$ satisfy $p^0\psi = f$. Apply the coboundary refinement relation (1): $p^1 d_{n+1}\psi = \frac{1}{2} d_n p^0 \psi = \frac{1}{2} d_n f$, which is fixed independently of ψ (subject to $p^0\psi = f$). The averaging-by-$\frac{1}{2}$ structure of p^1 on a 2-element fibre yields the calibration identity

$$p^1 (p^1)^* = \tfrac{1}{2}\, I_{X_n^1}, \tag{7}$$

which is verified directly: for any $\omega \in X_n^1$ and any level-n edge e with two children e_1, e_2, one has $((p^1)^*\omega)(e_i) = \frac{1}{2}\omega(e)$ (the unique adjoint under standard ℓ^2 inner products) and $(p^1(p^1)^*\omega)(e) = \frac{1}{2} \cdot \frac{1}{2}\omega(e) + \frac{1}{2} \cdot \frac{1}{2}\omega(e) = \frac{1}{2}\omega(e)$. Hence the orthogonal projection of any $\xi \in X_{n+1}^1$ onto $\text{range}((p^1)^*)$

is $P\xi = (p^1)^*\big((p^1(p^1)^*)^{-1}p^1\xi\big) = 2(p^1)^*p^1\xi$, and X^1_{n+1} decomposes orthogonally as $\mathrm{range}((p^1)^*) \oplus \ker p^1$. Apply this decomposition to $\xi = d_{n+1}\psi$:

$$d_{n+1}\psi = (p^1)^* d_n f \;+\; r_\psi, \qquad r_\psi \in \ker p^1, \tag{8}$$

where the first term comes from $P(d_{n+1}\psi) = 2(p^1)^*(p^1 d_{n+1}\psi) = 2(p^1)^*(\frac{1}{2}d_n f) = (p^1)^* d_n f$. Compute the norm using (7):

$$\|(p^1)^* d_n f\|^2 = \langle (p^1)^* d_n f, (p^1)^* d_n f\rangle = \langle d_n f, p^1(p^1)^* d_n f\rangle = \tfrac{1}{2}\|d_n f\|^2 = \tfrac{1}{2}\langle f, A^{\mathrm{vertex}}_n f\rangle.$$

By orthogonality of the decomposition (8),

$$\langle \psi, A^{\mathrm{vertex}}_{n+1}\psi\rangle = \|d_{n+1}\psi\|^2 = \tfrac{1}{2}\langle f, A^{\mathrm{vertex}}_n f\rangle + \|r_\psi\|^2 \;\geq\; \tfrac{1}{2}\langle f, A^{\mathrm{vertex}}_n f\rangle. \tag{9}$$

Step 3. We exhibit one specific ψ with $r_\psi = 0$, hence equality in (9): the harmonic extension itself. By the edge-midpoint hypothesis of the theorem statement, every interior vertex $m_e \in I$ has degree exactly 2 in $G^\bullet_{n+1}$, its only neighbours being the two endpoints u, v of its parent edge $e \in E(G^\bullet_n)$; hence the harmonic-extension formula $f_I = -A_{II}^{-1}A_{IB}f$ at m_e reduces to the elementary two-neighbour average $(H_{n+1}f)(m_e) = \frac{1}{2}(f(u) + f(v))$. Compute $d_{n+1}(H_{n+1}f)$ at the two child edges of e:

$$(d_{n+1}(H_{n+1}f))(u, m_e) = (H_{n+1}f)(m_e) - (H_{n+1}f)(u) = \tfrac{1}{2}(f(u)+f(v)) - f(u) = \tfrac{1}{2}(f(v)-f(u)) = \tfrac{1}{2}(d_n f)(e),$$
$$(d_{n+1}(H_{n+1}f))(m_e, v) = (H_{n+1}f)(v) - (H_{n+1}f)(m_e) = f(v) - \tfrac{1}{2}(f(u)+f(v)) = \tfrac{1}{2}(f(v)-f(u)) = \tfrac{1}{2}(d_n f)(e).$$

On the other hand, $((p^1)^* d_n f)(u, m_e) = \frac{1}{2}(d_n f)(e) = ((p^1)^* d_n f)(m_e, v)$ by definition of $(p^1)^*$ on a two-element fibre. Hence $d_{n+1}(H_{n+1}f) = (p^1)^* d_n f$ on every level-$(n+1)$ edge, i.e., $r_{H_{n+1}f} = 0$.

Combining Steps 1–3:

$$\langle f, \mathrm{Schur}(A^{\mathrm{vertex}}_{n+1})f\rangle = \langle H_{n+1}f, A^{\mathrm{vertex}}_{n+1}H_{n+1}f\rangle = \tfrac{1}{2}\langle f, A^{\mathrm{vertex}}_n f\rangle + \|r_{H_{n+1}f}\|^2 = \tfrac{1}{2}\langle f, A^{\mathrm{vertex}}_n f\rangle.$$

This holds for every $f \in X^0_n$. Since $\mathrm{Schur}(A^{\mathrm{vertex}}_{n+1})$ and A^{vertex}_n are both symmetric, polarisation of the quadratic form gives the operator identity (5). The scaled form (6) follows by homogeneity of degree one: the Schur complement is non-linear in blocks but $\mathrm{Schur}(c \cdot M) = c \cdot \mathrm{Schur}(M)$ for any scalar c, so $\mathrm{Schur}(\widetilde{A}_{n+1}) = \mathrm{Schur}(2^{n+1}A^{\mathrm{vertex}}_{n+1}) = 2^{n+1}\,\mathrm{Schur}(A^{\mathrm{vertex}}_{n+1}) = 2^{n+1}\cdot\frac{1}{2}A^{\mathrm{vertex}}_n = 2^n A^{\mathrm{vertex}}_n = \widetilde{A}_n$. □

Remark 2.2 (Numerical verification). The displayed Schur-equality and harmonic-compatibility matrix identities of Theorem 2.1 are verified to exact rational precision in §6 for two finite refinement instances: the single edge with one-midpoint subdivision (Tower I, $|V(G^\bullet_0)| = 2$, $|V(G^\bullet_1)| = 3$) and the triangle with edge-midpoint subdivision (Tower II, $|V(G^\bullet_0)| = 3$, $|V(G^\bullet_1)| = 6$). The numerical witness is `papers/cascade-refinement-compat/scripts/explore_defect.py`; the load-bearing matrix identities use exact `Fraction` arithmetic with no floating-point error. Eigenvalue diagnostics in the script are computed via floating-point `numpy.linalg.eigvalsh` as a sanity check, not as load-bearing verification.

3 Energy intertwining and discharge of (O3)

We now apply Theorem 2.1 to the cascade-mechanism consumer-API obligations. The cascade-mechanism residual functional in this paper's setting is $\widetilde{R}_n(\phi) := \frac{1}{2}\langle \phi, \widetilde{A}_n\phi\rangle$.

Theorem 3.1 (Energy intertwining; discharge of (O3)). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, for every $\psi \in X^0_{n+1}$,*

$$\widetilde{R}_n\big(p^0_{n+1,n}\psi\big) \;\le\; \widetilde{R}_{n+1}(\psi), \tag{10}$$

with equality if and only if $\psi = H_{n+1}\big(p^0_{n+1,n}\psi\big)$ (i.e. ψ is the harmonic extension of its boundary).

Proof. Decompose $\psi = H_{n+1}(p^0_{n+1,n}\psi) + \psi^\perp$, where $\psi^\perp := \psi - H_{n+1}(p^0_{n+1,n}\psi)$ vanishes on $B = V(G^\bullet_n)$ (since $p^0_{n+1,n}H_{n+1}(p^0_{n+1,n}\psi) = p^0_{n+1,n}\psi$). The energy quadratic form decomposes as

$$\begin{aligned}\langle\psi, A^{\text{vertex}}_{n+1}\psi\rangle = {}&\langle H_{n+1}(p^0\psi), A^{\text{vertex}}_{n+1}H_{n+1}(p^0\psi)\rangle\\ &+ 2\langle H_{n+1}(p^0\psi), A^{\text{vertex}}_{n+1}\psi^\perp\rangle + \langle\psi^\perp, A^{\text{vertex}}_{n+1}\psi^\perp\rangle.\end{aligned}$$

The cross term vanishes: writing A^{vertex}_{n+1} in block form (3) and noting that $\psi^\perp$ has zero B-component, one has $A^{\text{vertex}}_{n+1}\psi^\perp = (A_{BI}\psi^\perp_I, A_{II}\psi^\perp_I)$, and $H_{n+1}(p^0\psi) = (p^0\psi, -A^{-1}_{II}A_{IB}p^0\psi)$; the cross-pairing computes to $\langle p^0\psi, A_{BI}\psi^\perp_I\rangle + \langle -A^{-1}_{II}A_{IB}p^0\psi, A_{II}\psi^\perp_I\rangle = \langle p^0\psi, A_{BI}\psi^\perp_I\rangle - \langle p^0\psi, A_{BI}\psi^\perp_I\rangle = 0$. The first surviving term equals $\langle p^0\psi, \mathrm{Schur}(A^{\text{vertex}}_{n+1})p^0\psi\rangle = \frac{1}{2}\langle p^0\psi, A^{\text{vertex}}_n p^0\psi\rangle$ by Theorem 2.1. The third surviving term is $\langle\psi^\perp_I, A_{II}\psi^\perp_I\rangle \ge 0$ since A_{II} is positive semi-definite (a principal submatrix of the positive-semi-definite A^{vertex}_{n+1}). Hence

$$\langle\psi, A^{\text{vertex}}_{n+1}\psi\rangle \;\ge\; \tfrac{1}{2}\langle p^0\psi, A^{\text{vertex}}_n p^0\psi\rangle.$$

Multiplying both sides by $\frac{1}{2}\cdot 2^{n+1}$ and regrouping factors using $\widetilde{A}_{n+1} = 2^{n+1}A^{\text{vertex}}_{n+1}$ and $\widetilde{A}_n = 2^n A^{\text{vertex}}_n$:

$$\tfrac{1}{2}\langle\psi, \widetilde{A}_{n+1}\psi\rangle \;\ge\; \tfrac{1}{2}\langle p^0\psi, \widetilde{A}_n p^0\psi\rangle,$$

i.e. $\widetilde{R}_{n+1}(\psi) \ge \widetilde{R}_n(p^0\psi)$, which is (10). Equality requires $\psi^\perp = 0$ (since A_{II} is positive definite by invertibility), i.e. ψ is the harmonic extension of its boundary. □

Corollary 3.2 ((O3) is closed in the abstract pure-midpoint model). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, the cascade-mechanism residual-monotonicity hypothesis [2] (O3) holds for the scaled curvature residual $\widetilde{R}_n(\phi) = \frac{1}{2}\langle\phi, \widetilde{A}_n\phi\rangle$, where $\widetilde{A}_n = 2^n A^{\text{vertex}}_n$ and $A^{\text{vertex}}_n = d^*_n d_n$ is the vertex graph Laplacian. Lifting this discharge to the full Schläfli P3/P4 tower of [4, 3] (which adds face-centroid vertices and centroid–midpoint edges absent from Definition 1.1) is open and named in §1.1.*

4 Strict operator-level refinement-compatibility fails

Theorem 2.1 closes the energy-level refinement question in the abstract model. The strict operator-level identity $p^0_{n+1,n}\widetilde{A}_{n+1} = \widetilde{A}_n p^0_{n+1,n}$ remains false, with explicit defect.

Proposition 4.1 (Strict operator intertwining fails; kernel of the defect). *In the abstract scalar pure-midpoint refinement model of Definition 1.1, the strict identity $p^0_{n+1,n}\widetilde{A}_{n+1} = \widetilde{A}_n p^0_{n+1,n}$ does not hold in general, and equivalently $p^0_{n+1,n}A^{\text{vertex}}_{n+1} \ne \frac{1}{2}A^{\text{vertex}}_n p^0_{n+1,n}$ as operators $X^0_{n+1} \to X^0_n$. The defect operator*

$$D_n := p^0_{n+1,n}A^{\text{vertex}}_{n+1} - \tfrac{1}{2}A^{\text{vertex}}_n p^0_{n+1,n} \tag{11}$$

satisfies the following:

(i) *$D_n\psi = 0$ whenever $\psi \in \mathrm{range}(H_{n+1})$, i.e. whenever ψ is the harmonic extension of its own boundary $p^0\psi$;*

(ii) *Writing* $\delta(\psi) \in \mathbb{R}^{E_n}$ *for the per-edge midpoint deficit* $\delta_e(\psi) := \frac{1}{2}(\psi_u + \psi_v) - \psi_{m_e}$ *(with* $e = \{u,v\} \in E_n$ *and* m_e *the midpoint inserted on* e*), one has* $(D_n\psi)_v = \sum_{e \ni v} \delta_e(\psi)$ *for every* $v \in V_n$*. Equivalently,* $D_n\psi = \mathcal{B}_n\delta(\psi)$ *where* $\mathcal{B}_n \in \{0,1\}^{V_n \times E_n}$ *is the unsigned vertex–edge incidence matrix of* G_n *(the calligraphic* $\mathcal{B}_n$ *disambiguates from the coboundary block* B_n *of [3] Definition* `def:Bn`*, which is a different operator). Consequently*

$$\ker D_n \;=\; \mathrm{range}(H_{n+1}) + \bigl\{\psi : \delta(\psi) \in \ker \mathcal{B}_n,\; \psi_B = 0\bigr\}, \tag{12}$$

with the two summands intersecting trivially, so $\ker D_n = \mathrm{range}(H_{n+1})$ *if and only if* $\ker \mathcal{B}_n = 0$*. Standard spectral graph theory identifies* $\ker \mathcal{B}_n \subseteq \mathbb{R}^{E_n}$ *via the rank formula* $\mathrm{rank}(\mathcal{B}_n) = |V_n| - b(G_n)$*, where* $b(G_n)$ *is the number of bipartite connected components of* G_n *(equivalently, the signless Laplacian* $Q(G_n) := \mathcal{B}_n\mathcal{B}_n^T = D(G_n) + A(G_n)$ *acting on* $\mathbb{R}^{V_n}$ *has* $\dim\ker Q(G_n) = b(G_n)$*, and* $\mathrm{rank}(\mathcal{B}_n) = \mathrm{rank}(\mathcal{B}_n\mathcal{B}_n^T)$*). Hence* $\dim\ker\mathcal{B}_n = |E_n| - |V_n| + b(G_n)$*, and in particular* $\ker\mathcal{B}_n = 0$ *whenever* G_n *is a (simple) forest. The smallest simple-graph counterexample is* $G_n = C_4$ *(the 4-cycle), where* $\dim\ker\mathcal{B}_n = 1$ *and* $\ker D_n$ *is strictly larger than* $\mathrm{range}(H_{n+1})$ *(a parallel-edge multigraph already has* $\ker\mathcal{B}_n \neq 0$*, but Definition 1.1 restricts to simple* G_n*);*

(iii) *the restriction* $D_n|_{\{\psi:\psi_I=0\}}$ *is not the zero operator whenever* G_n *has at least one non-isolated vertex. Concretely, the constant-on-boundary state* $\psi_B \equiv 1$*,* $\psi_I \equiv 0$ *has* $\delta_e(\psi) = 1$ *on every edge, so* $(D_n\psi)_v = \deg_{G_n}(v) > 0$ *at every non-isolated* v*.*

Proof. **Per-vertex formula.** Fix $v \in V_n$. By Definition 1.1 (M2), the level-$(n+1)$ neighbours of v are exactly the midpoints $\{m_e : e \in E_n,\, e \ni v\}$ of the level-n edges incident to v, and v has the same level-$(n+1)$ degree $\deg_{G_{n+1}}(v) = \deg_{G_n}(v)$. Hence

$$(A_{n+1}^{\mathrm{vertex}}\psi)(v) \;=\; \sum_{e \ni v}\bigl(\psi(v) - \psi(m_e)\bigr) \quad \text{and} \quad (A_n^{\mathrm{vertex}}\psi_B)(v) \;=\; \sum_{e=\{v,u\}\ni v}\bigl(\psi(v) - \psi(u)\bigr),$$

so for every $\psi \in X_{n+1}^0$ with $\psi_B = p^0\psi$,

$$\begin{aligned}
(D_n\psi)(v) &= (A_{n+1}^{\mathrm{vertex}}\psi)(v) - \tfrac{1}{2}(A_n^{\mathrm{vertex}}\psi_B)(v) \\
&= \sum_{e \ni v}\bigl(\psi(v) - \psi(m_e)\bigr) - \tfrac{1}{2}\sum_{e=\{v,u\}}\bigl(\psi(v) - \psi(u)\bigr) \\
&= \sum_{e \ni v}\Bigl[\tfrac{1}{2}(\psi(u) + \psi(v)) - \psi(m_e)\Bigr] \;=\; \sum_{e \ni v}\delta_e(\psi) \;=\; (\mathcal{B}_n\delta(\psi))(v).
\end{aligned}$$

This proves the formula in the proposition statement.

(i) If $\psi = H_{n+1}f$ is the harmonic extension of $f \in X_n^0$, the harmonic-equation $(A_{II})_{m_e,m_e}\psi(m_e) + (A_{IB})_{m_e,\cdot}f = 0$ collapses, since m_e has level-$(n+1)$ neighbours $\{u,v\}$ (so A_{II} is diagonal with entries 2 on midpoints, A_{IB} has rows $-e_u^T - e_v^T$ on midpoint m_e), to $2\psi(m_e) = f(u) + f(v)$, i.e. $\psi(m_e) = \frac{1}{2}(f(u) + f(v))$. Hence $\delta_e(\psi) = \frac{1}{2}(f(u) + f(v)) - \psi(m_e) = 0$ for all $e \in E_n$, and the per-vertex formula gives $D_n\psi = \mathcal{B}_n \cdot 0 = 0$.

(ii) From the per-vertex formula above, $D_n\psi = 0$ iff $\mathcal{B}_n\delta(\psi) = 0$ iff $\delta(\psi) \in \ker\mathcal{B}_n$. The map $\psi \mapsto \delta(\psi)$ is surjective $X_{n+1}^0 \twoheadrightarrow \mathbb{R}^{E_n}$ (any prescribed $\delta \in \mathbb{R}^{E_n}$ is realised by $\psi_B := 0$ and $\psi_{m_e} := -\delta_e$), and its kernel is exactly $\mathrm{range}(H_{n+1})$ by the harmonic-extension equation $\psi(m_e) = \frac{1}{2}(\psi(u) + \psi(v))$ in part (i). Hence the short exact sequence $0 \to \mathrm{range}(H_{n+1}) \to \ker D_n \to \ker\mathcal{B}_n \to 0$ splits via the section $\delta \mapsto \psi$ defined by $\psi_B = 0$, $\psi_{m_e} = -\delta_e$, giving (12). The dimension formula $\dim\ker\mathcal{B}_n = |E_n| - |V_n| + b(G_n)$, where $b(G_n)$ counts bipartite connected components, follows from $\mathrm{rank}(\mathcal{B}_n) = |V_n| - b(G_n)$ via the signless-Laplacian identity $\mathcal{B}_n\mathcal{B}_n^T = D(G_n) + A(G_n)$ and the standard fact that $\dim\ker(D+A) = b(G_n)$ on a graph (e.g. [1, Prop. 1.3.10]). For $G_n = C_4$ this gives $\dim\ker\mathcal{B}_n = 4-4+1 = 1$, with explicit generator $\delta = (1,-1,1,-1)$ on the four edges in cyclic order.

(iii) For any $\phi \in X^0_{n+1}$ with $\phi_I = 0$ and edge $e = \{u,v\} \in E_n$, $\delta_e(\phi) = \frac{1}{2}(\phi(u)+\phi(v))$. Take the constant-on-boundary witness $\phi_B \equiv 1$, $\phi_I \equiv 0$: every $\delta_e(\phi) = 1$, so the per-vertex formula gives $(D_n\phi)(v) = (\mathcal{B}_n\delta(\phi))(v) = \deg_{G_n}(v)$, which is strictly positive at every non-isolated vertex. Hence $D_n|_{\{\phi_I=0\}}$ is not the zero operator whenever G_n has at least one non-isolated vertex. The explicit defect $D_n\phi$ is computed for two concrete towers in §6; in both, $D_n|_{\{\phi_I=0\}}$ is non-zero. □

Remark 4.2 (Geometric interpretation). The conclusion (i) of Proposition 4.1 is the operator-level shadow of Theorem 2.1: harmonic extensions saturate the boundary Laplacian action under p^0 at the strict scaled level. Conclusion (ii) records the genuinely finer fact that, on parent graphs G_n with non-trivial unsigned incidence kernel (the smallest example is $G_n = C_4$), the kernel of D_n is strictly larger than $\mathrm{range}(H_{n+1})$: there exist non-harmonic level-$(n+1)$ states whose midpoint deficits form a non-zero edge-cocycle in $\ker\mathcal{B}_n$ but whose total at every boundary vertex still cancels. The defect D_n measures the boundary-vertex sum of midpoint deficits $\delta_e(\psi) = \frac{1}{2}(\psi_u+\psi_v) - \psi_{m_e}$ along incident edges, not the per-edge midpoint deficit itself.

Remark 4.3 (Energy vs. flow level). The Schur identity Theorem 2.1 and its energy consequence Theorem 3.1 hold at the *quadratic-form / boundary-energy* level. Strict *operator*-level intertwining $p^0\widetilde{A}_{n+1} = \widetilde{A}_n p^0$ fails in general by Proposition 4.1, and consequently the strict *flow*-level intertwining $p^0 \circ e^{-t\widetilde{A}_{n+1}} = e^{-t\widetilde{A}_n} \circ p^0$ also fails in general: differentiating any such semigroup identity at $t=0$ would recover the false operator identity (the edgeless / trivial case where both operators vanish is the degenerate exception). Cascade-mechanism's obligation (O2), as stated, is at the flow level. A useful boundary-energy substitute adjacent to (O2) is the Schur-energy identity $\mathrm{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$, which Theorem 2.1 discharges. In the cascade-mechanism reformulation table of §7 we record (O2) *as stated* as still false, with the Schur-energy substitute discharged.

5 Kernel of A_n^{vertex} and (O1)

For completeness we record the status of cascade-mechanism's obligation (O1) (top-rung residual-zero fixed point and convergence) in the abstract scalar pure-midpoint refinement model of Definition 1.1, applied at the top level $n = N$.

Proposition 5.1 (Kernel and convergence of $\widetilde{A}_N$)**.** *For any finite level N, $\widetilde{A}_N$ is positive semi-definite on X^0_N and its kernel $\ker\widetilde{A}_N = \ker A_N^{\mathrm{vertex}}$ consists exactly of the locally-constant functions on the connected components of $G^\bullet_N$. If $G^\bullet_N$ is connected,* $\dim\ker\widetilde{A}_N = 1$ *(constants only).*

For every initial datum $\phi^{(0)} \in X^0_N$, the gradient flow $\Psi^t_N := e^{-t\widetilde{A}_N}$ converges as $t\to\infty$ to the orthogonal projection of $\phi^{(0)}$ onto $\ker\widetilde{A}_N$. This is automatic in finite dimensions by spectral decomposition.

In particular, the trivial state $\phi^\star_N = 0$ is always a residual-zero fixed point. (O1) as a nontrivial-*selection obligation does not collapse to whether $\ker\widetilde{A}_N \neq \{0\}$: for any nonempty graph the constants already give $\ker\widetilde{A}_N \neq \{0\}$. The real content of (O1) is the choice of a specific nontrivial element of $\ker\widetilde{A}_N$ (or equivalently a nonzero projection of the initial datum onto that kernel), and this selection is out of scope per [2] §3.4.*

Proof. $A_N^{\mathrm{vertex}} = d_N^* d_N$ is positive semi-definite by construction, with kernel $\ker d_N$, the locally-constant functions on the connected components of $G^\bullet_N$. The convergence statement for $e^{-tA_N^{\mathrm{vertex}}}$

is the standard finite-dimensional functional-calculus result for a self-adjoint positive-semi-definite operator. Multiplying by the positive scalar 2^N does not change the kernel or the convergence behaviour. $\square$

Remark 5.2 (Selection rule is not part of cascade closure). [2] §3.4 explicitly states that selection of the top-rung fixed point is not provided by cascade closure and is the successor paper's own task. The kernel description above records what cascade closure inherits at the top rung; the selection rule by which a particular non-zero element of $\ker \widetilde{A}_N$ is preferred over the zero state is a question for downstream papers, and outside the scope of the present source-side discharge.

6 Worked finite refinement towers

This section gives explicit matrix-level computations on two concrete finite refinement towers, verifying the load-bearing identities of Theorem 2.1 (Schur halving) and the defect formula of Proposition 4.1 to exact rational precision. Theorem 3.1 is an analytic corollary of Theorem 2.1 valid on every ψ, not a finite computation, and is not directly numerically witnessed; the identities verified below are the inputs to that analytic step. The numerical witness is the script `papers/cascade-refinement-compat/scripts/explore_defect.py`; its output is preserved in `papers/cascade-refinement-compat/repro/expected_outputs/`.

6.1 Tower I: single edge → one-midpoint subdivision

Let $G_0^\bullet$ be the graph on $V(G_0^\bullet) = \{u, v\}$ with the single edge $e = (u, v)$. Refine by inserting the midpoint m of e, splitting e into two new edges (u, m) and (m, v): $V(G_1^\bullet) = \{u, m, v\}$, $E(G_1^\bullet) = \{(u, m), (m, v)\}$. The discrete coboundaries in oriented form are $d_0 = \begin{pmatrix} -1 & 1 \end{pmatrix}$ on $X_0^0 = \mathbb{R}^2$, and $d_1 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ on $X_1^0 = \mathbb{R}^3$. The bonding maps are $p_{1,0}^0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $p_{1,0}^1 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \end{pmatrix}$. The vertex Laplacians are

$$A_0^{\text{vertex}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad A_1^{\text{vertex}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

With the partition $B = \{u, v\}$, $I = \{m\}$, the block decomposition yields $A_{BB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A_{BI} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$, $A_{IB} = \begin{pmatrix} -1 & -1 \end{pmatrix}$, $A_{II} = \begin{pmatrix} 2 \end{pmatrix}$. The Schur complement is

$$\mathrm{Schur}(A_1^{\text{vertex}}) = A_{BB} - A_{BI} A_{II}^{-1} A_{IB} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \tfrac{1}{2} A_0^{\text{vertex}}.$$

This is exactly Theorem 2.1.

The defect of strict refinement-compatibility is

$$D_0 := p_{1,0}^0 A_1^{\text{vertex}} - \tfrac{1}{2} A_0^{\text{vertex}} p_{1,0}^0 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -1 & \frac{1}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} \end{pmatrix}.$$

Both rows of D_0 act on $\phi = (\phi(u), \phi(m), \phi(v))$ as $\frac{1}{2}\phi(u) - \phi(m) + \frac{1}{2}\phi(v) = \frac{1}{2}(\phi(u) + \phi(v)) - \phi(m)$, the "midpoint deficit" of ϕ at m. Tower I has a single edge with no cycles, so the unsigned vertex–edge incidence matrix $\mathcal{B}_0 \in \{0,1\}^{2\times 1}$ has trivial kernel, and the incidence characterisation in Proposition 4.1(ii) collapses to: $D_0\phi = 0$ iff the midpoint deficit at m is zero, iff $\phi(m) = \frac{1}{2}(\phi(u)+\phi(v))$, iff ϕ is the harmonic extension of its boundary. So here the kernel of D_0 coincides with range(H_1). Furthermore, D_0 does *not* vanish on $\{\phi : \phi(m) = 0\}$ (which gives $D_0\phi = (\frac{1}{2}(\phi(u)+\phi(v)), \frac{1}{2}(\phi(u)+\phi(v)))$, generically non-zero), confirming Proposition 4.1(iii).

6.2 Tower II: triangle → edge-midpoint subdivision

Let $G_0^\bullet$ be the 3-cycle on $V(G_0^\bullet) = \{0,1,2\}$ with edges $e_{01} = (0,1)$, $e_{12} = (1,2)$, $e_{20} = (2,0)$. Refine by inserting one midpoint per edge, m_{01}, m_{12}, m_{20}, giving $V(G_1^\bullet) = \{0,1,2,m_{01},m_{12},m_{20}\}$ ($|V(G_1^\bullet)| = 6$) and the six new edges $(0,m_{01}),(m_{01},1),(1,m_{12}),(m_{12},2),(2,m_{20}),(m_{20},0)$. With $B = \{0,1,2\}$, $I = \{m_{01},m_{12},m_{20}\}$ and the identification of bonding maps $p^0_{1,0}$, $p^1_{1,0}$ analogously to Tower I, direct computation (verified by `explore_defect.py`) yields

$$A_0^{\mathrm{vertex}} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

and the Schur complement

$$\mathrm{Schur}(A_1^{\mathrm{vertex}}) = \begin{pmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{pmatrix} = \tfrac{1}{2}A_0^{\mathrm{vertex}},$$

again confirming Theorem 2.1. Eigenvalues of the *Schur difference* $\mathrm{Schur}(A_1^{\mathrm{vertex}}) - A_0^{\mathrm{vertex}}$ (distinct from the named defect D_n of Proposition 4.1) on this tower are $\{0, -\frac{3}{2}, -\frac{3}{2}\}$, confirming the ordering $\mathrm{Schur}(A_1^{\mathrm{vertex}}) \le A_0^{\mathrm{vertex}}$ (i.e. the unscaled Schur complement under-estimates the level-0 Laplacian; rescaling by 2^n at level n exactly compensates).

The defect operator D_0 of Proposition 4.1 on Tower II, with column ordering $(\phi(0), \phi(1), \phi(2), \phi(m_{01}), \phi(m_{12}), \phi($ is the 3×6 matrix

$$D_0 = p^0_{1,0}A_1^{\mathrm{vertex}} - \tfrac{1}{2}A_0^{\mathrm{vertex}}p^0_{1,0} = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & -1 & 0 & -1 \\ \frac{1}{2} & 1 & \frac{1}{2} & -1 & -1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 & 0 & -1 & -1 \end{pmatrix}.$$

On the constant-on-boundary witness $\phi_B = (1,1,1)$, $\phi_I = 0$ from the proof of Proposition 4.1(iii), $D_0\phi = (2,2,2) = (\deg_{G_0}(0), \deg_{G_0}(1), \deg_{G_0}(2))$, in agreement with the per-vertex formula. (Verified to exact `Fraction` arithmetic by the script.)

7 Cascade-mechanism update

The present paper updates the worked-invocation status table of [2] §3.5 in the abstract scalar pure-midpoint refinement model of Definition 1.1, which §1.1 relates to the imported P3/P4 finite-level refinement tower of [4, 3]: an update lifts to that tower under the named hypotheses (L1) and (L3) of §1.1 ((L2) is already a theorem of P3) and otherwise stays inside the abstract model. The consumer-API obligations themselves ([2] §3.4) are unchanged; the cascade-closure mechanism, Definition, Proposition, and Corollary of [2] are unchanged.

Table 1: Updated status of cascade-mechanism consumer-API obligations in the abstract scalar pure-midpoint refinement model (Definition 1.1) with the scaled curvature operator $\widetilde{A}_n = 2^n A_n^{\mathrm{vertex}}$. (O0) is immediate in the abstract finite-dimensional linear model: finite spaces and bonding maps come from [4], and flow well-posedness is elementary finite-dimensional matrix theory. the (O3) row and the boundary-energy (O2) substitute lift to the full P3/P4 tower of [4, 3] under the open hypotheses (L1) and (L3) of §1.1 (the third lift hypothesis (L2) is already a theorem of [4]).

Obligation	**Status (this paper)**	**Where established**
(O0) Standing analytic / admissibility hypotheses	**Closed.** Finite-dimensional Hilbert structure of X_n^0, C^1 bonding maps, well-posed flow $e^{-t\widetilde{A}_n}$.	[4] for finite spaces / bonding maps; well-posedness of $e^{-t\widetilde{A}_n}$ is an elementary finite-matrix fact here. Recap in §1.
(O1) Top-rung fixed point and convergence	**Trivial fixed point closed** (zero state is always residual-zero, and $e^{-t\widetilde{A}_N}$ converges to the orthogonal projection onto $\ker \widetilde{A}_N$ on every component). **Nontrivial selection of an element of** $\ker \widetilde{A}_N \neq \{0\}$ **remains out of scope** per [2] §3.4 (it is the successor paper's task).	Proposition 5.1.
(O2) Flow intertwining	**(O2) as stated remains false in general.** Strict operator-level identity $p^0 \widetilde{A}_{n+1} = \widetilde{A}_n p^0$ fails for nontrivial refinements (Proposition 4.1), so the flow-level identity $p^0 \circ e^{-t\widetilde{A}_{n+1}} = e^{-t\widetilde{A}_n} \circ p^0$ also fails in general. The boundary-energy / Schur substitute holds exactly: $\mathrm{Schur}(\widetilde{A}_{n+1}) = \widetilde{A}_n$ (Theorem 2.1). This proves the substitute; it does not discharge (O2) as stated.	Theorem 2.1, Proposition 4.1.
(O3) Residual monotonicity	**Closed.** $\widetilde{R}_n(p^0\psi) \leq \widetilde{R}_{n+1}(\psi)$ for every $\psi \in X_{n+1}^0$, with equality iff ψ is the harmonic extension of its boundary.	Theorem 3.1, Corollary 3.2.

The discharge of (O3) and the boundary-energy / Schur substitute for (O2) are non-trivial relative to [2]'s original status report; the structural discovery is that, in the abstract scalar model, the source's coboundary factor-$\frac{1}{2}$ relation (1) has a curvature-level Schur analogue, with the scaled operator $\widetilde{A}_n := 2^n A_n^{\mathrm{vertex}}$ as the natural refinement-compatible object at the boundary-energy level.

8 Discussion

What this paper does. It proves that, in the abstract scalar pure-midpoint refinement model of Definition 1.1 and with the scaled curvature operator $\widetilde{A}_n = 2^n A_n^{\text{vertex}}$, the cascade-mechanism residual-monotonicity obligation (O3) holds (Theorem 3.1, Corollary 3.2). It also discharges a boundary-energy / Schur substitute for the flow-intertwining obligation (O2) (Theorem 2.1); (O2) *as stated*, at the operator/flow level, remains false (Proposition 4.1). The mechanism by which both happen is the same: the coboundary factor-$\frac{1}{2}$ relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$, which holds verbatim in the abstract model (and as a theorem in the source's full Schläfli setting), pulled forward through the boundary–interior decomposition.

What this paper does not do. It does not establish strict operator-level $p^0 \widetilde{A}_{n+1} = \widetilde{A}_n p^0$ intertwining; that identity is genuinely false in the abstract model. The defect operator D_n defined in Proposition 4.1 satisfies $\operatorname{range}(H_{n+1}) \subseteq \ker D_n$, with equality exactly when $\ker \mathcal{B}_n = 0$ (e.g. when G_n is a forest), and D_n is non-zero on the level-n-supported sector $\{\psi : \psi_I = 0\}$ in general. In particular, the strict cascade-mechanism flow-intertwining hypothesis (O2) at the operator/flow level remains a strictly stronger condition than what the abstract model supports; cascade-mechanism users invoking the boundary-energy version of (O2) will find it discharged here, but those requiring strict operator-level intertwining will find the defect computed but not zero. The paper also does not lift any of these results to the full Schläfli P3/P4 tower of [4, 3]: that lift is named in §1.1 as hypotheses (L1)–(L3) and is not attempted here. The paper does not derive a selection rule for nontrivial elements of $\ker \widetilde{A}_N$, in keeping with [2] §3.4's explicit boundary.

Implication for cascade-mechanism. The cascade-mechanism worked-invocation status table ([2] §3.5) has already been updated to incorporate this paper's discharge: (O3) closed for the scaled curvature $\widetilde{A}_n$ via Theorem 3.1 and Corollary 3.2 (using Theorem 2.1) in the abstract pure-midpoint model, with the P3/P4-tower lift flagged conditional on the lift hypotheses of §1.1 (in particular (L1) and (L3) for the (O3) row, since (L2) is already a theorem of P3, see §1.1), and the boundary-energy / Schur substitute for (O2) discharged in the same scope. The cascade-mechanism consumer API itself ([2] §3.4 (O0)–(O3)) does not change.

Open follow-ups. Proposition 4.1(ii) characterises $\ker D_n$ via the unsigned vertex–edge incidence operator $\mathcal{B}_n$. The natural follow-up is the structure of $\ker D_n \setminus \operatorname{range}(H_{n+1})$ when G_n has non-trivial $\ker \mathcal{B}_n$ (e.g. $G_n = C_4$), and whether such "incidence-cocycle" kernel directions form an invariant sector under the cascade-mechanism flow $e^{-t\widetilde{A}_{n+1}}$. A separate question is whether a non-linear cascade-flow modification (gradient flow on a modified residual that absorbs the defect into the dynamics) preserves the mechanism's structure while admitting strict intertwining; this is out of scope for the present paper. The P3/P4-tower lift under (L1)–(L3) is the most load-bearing follow-up.

9 Reproducibility

The displayed Schur and defect matrix equalities in Sections 2–6 are verified to exact rational precision (`Fraction` arithmetic) by the script `papers/cascade-refinement-compat/scripts/explore_defect.py`. Eigenvalue diagnostics in the script are computed via floating-point `numpy.linalg.eigvalsh` as a sanity check; the load-bearing equalities are the exact `Fraction`-level matrix identities, not the dis-

played eigenvalues. A reproducer with frozen reference outputs lives at `papers/cascade-refinement-compat/repr`
from the repository root, one invocation

```
cd papers/cascade-refinement-compat/repro
bash run_audit.sh
```

re-runs the verification on Towers I and II and diffs against the frozen expected outputs. A clean run prints `REFINEMENT COMPAT AUDIT PASSED`.

References

[1] Andries E. Brouwer and Willem H. Haemers. *Spectra of Graphs*. Universitext. Springer, 2012.

[2] Lee Smart. Cascade closure: A conditional compatibility template for geometric state selection. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-mechanism/cascade-mechanism.tex`. The consumer API (O0)–(O3) discharged in part by the present paper is in `cascade-mechanism.tex` §3.4., 2026.

[3] Lee Smart. Cascade closure dynamics: Finite closure functionals, gradient flow, and refinement compatibility. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-closure-dynamics/cascade-closure-dynamics.tex`. The coboundary-refinement relation $p^1 d_{n+1} = \frac{1}{2} d_n p^0$ used as the load-bearing source identity is Proposition `prop:coboundary-refinement` of that paper., 2026.

[4] Lee Smart. Cascade refinement spaces: Finite-level state spaces and bonding maps. Preprint, Institute of Vibrational Field Dynamics. Local source: `papers/cascade-refinement-spaces/cascade-refinement-spaces.tex`., 2026.